

- Q.26 Explain Fire clay.
 Q.27 Explain Quartzite.
 Q.28 Explain Permanent linear change.
 Q.29 Explain phase diagram.
 Q.30 List the composition of chromitite refractory.
 Q.31 List the uses of carbon refractory.
 Q.32 Differentiate between true porosity and apparent porosity.
 Q.33 Explain Mullite.
 Q.34 Discuss Permeability.
 Q.35 Explain Magnesium-Chrome refractories.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 List the factors affecting of selection of refractories raw material. Describe occurrence of raw material and manufacturing units in India.
 Q.37 Describe the MgO-SiO₂ phase diagram with help of neat sketch.
 Q.38 Describe the testing method of determination of porosity of a given sample of refractory.

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4th Semester/ Ceramic Subject : Ceramic Refractory Technology-I

Time : 3 Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Example of acidic refractory is _____.
 a) Dolomite refractory
 b) Quartz refractory
 c) Zirconia refractory
 d) Mag-Chrom refractory
- Q.2 Examples of Neutral refractory is _____.
 a) Fire clay refractory
 b) Silica refractory
 c) Magnesite refractory
 d) Carbon refractory
- Q.3 _____ is the volume of air or gas which will pass through a cubic centimetre of the material under a pressure of 1 cm of water in one second
 a) Permeability b) PCE
 c) CCS d) Bulk Density

- Q.4 The refractory are neither attacked by acid slag nor by basics lag is called
- a) Acid refractory b) Basic refractory
c) Neutral refractory d) None
- Q.5 PCE stands for ____.
- a) Pyrometric cone equivalent
b) Pyrometric cylinder equivalent
c) Pyrometric card equivalent
d) Pyrometric care equivalent
- Q.6 Silicon Carbide has chemical formula
- a) CaO b) SiB
c) SiC d) HCl
- Q.7 In PCE est the shape of sample cone is
- a) Prism b) Tetrahedron
c) Square d) Sphere
- Q.8 The B.D. Stands for
- a) Bend Density b) Bulk density
c) Density d) None
- Q.9 Sillimanite occurs in Travan core in
- a) Kerala b) Bihar
c) Jharkhand d) Haryana
- Q.10 The porosity of insulating refractory is
- a) 80% b) 10-12%
c) 20% d) 2-5%

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SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (10x1=10)

- Q.11 Weight per unit volume of refractory is known as ____.
- Q.12 Silica refractories are basic in nature. (True/False)
- Q.13 Firing of refractories are done is tunnel kiln. (True/False)
- Q.14 Capacity of a refractory brick to resist the rubbing action by moving charge is called as ____.
- Q.15 ____ is a acidic refractory.
- Q.16 RUL test determines the ____ of refractories.
- Q.17 Silica content in silica refractory can be as high as ____ percent.
- Q.18 Price of refractory brick should be low. (True/False)
- Q.19 Slag resistance is one of the important properties of refractory. (True/False)
- Q.20 Temperature inside the kiln may be measured with the help of thermometer. (True/False)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Classify the fire clay refractories
- Q.22 Explain density.
- Q.23 Explain basic refractories.
- Q.24 Discuss preparation of chromite refractory.
- Q.25 Explain fusion cast refractories.

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